Advanced Python for Data Science (TRN001)

1. Course Overview

- **Duration:** 3 Days | 4 Hours per Day | Total 12 Hours
- **Mode:** Virtual Instructor-Led Training (VILT)
- **Level:** Intermediate to Advanced
- **Target Audience:** Data Analysts, Data Scientists, AI Engineers, ML Engineers

2. Learning Outcomes

- Write efficient and maintainable Python code using advanced programming practices.
- Process and transform enterprise-scale datasets using Pandas.
- Apply statistical analysis and hypothesis testing for business decisions.
- Build, evaluate, and optimize machine learning models.
- Understand deployment strategies for real-world AI solutions.
- Solve business problems using analytics and predictive modeling.

3. Curriculum Structure

Day 1 – Advanced Python & Data Processing

Session	Session 1: Advanced Python Programming
Topics Covered	Python Best Practices, OOP, SOLID Principles, Design Patterns, Type Hinting
Live Demonstration / Case Study	Build reusable analytics utility package
Hands-On Activity	Create reusable classes for data processing
Outcome	Participants gain practical implementation experience

Session	Session 2: Advanced Python Features
Topics Covered	Iterators, Generators, Decorators, Context Managers, Exception Handling
Live Demonstration / Case Study	Optimize dataset processing using generators
Hands-On Activity	Refactor existing code for better performance
Outcome	Participants gain practical implementation experience

Session	Session 3: Data Manipulation using Pandas
Topics Covered	Advanced DataFrames, Merge & Join, GroupBy, Window Functions, Pivot Tables
Live Demonstration / Case Study	Sales Performance Analytics
Hands-On Activity	Build KPI calculations from raw datasets
Outcome	Participants gain practical implementation experience

Session	Session 4: Data Cleaning Workshop
Topics Covered	Missing Values, Duplicate Handling, Outlier Detection, Validation
Live Demonstration / Case Study	Production-ready dataset preparation
Hands-On Activity	Clean and validate datasets
Outcome	Participants gain practical implementation experience

Day 2 – Statistics & Machine Learning Essentials

Session	Session 5: Statistics for Data Science
Topics Covered	Descriptive Statistics, Probability, Distributions, Correlation
Live Demonstration / Case Study	Customer Behavior Analysis
Hands-On Activity	Interpret statistical insights
Outcome	Participants gain practical implementation experience

Session	Session 6: Hypothesis Testing & A/B Testing
Topics Covered	T-Test, Chi-Square, Experiment Design, A/B Testing
Live Demonstration / Case Study	Website Conversion Optimization
Hands-On Activity	Design testing strategy
Outcome	Participants gain practical implementation experience

Session	Session 7: Feature Engineering
Topics Covered	Feature Creation, Selection, Encoding, Scaling
Live Demonstration / Case Study	ML Dataset Preparation
Hands-On Activity	Engineer predictive features
Outcome	Participants gain practical implementation experience

Session	Session 8: Machine Learning Fundamentals
Topics Covered	Regression, Classification, Clustering, Model Selection
Live Demonstration / Case Study	Customer Churn Prediction
Hands-On Activity	Build basic ML models
Outcome	Participants gain practical implementation experience

Day 3 – Model Building & Deployment

Session	Session 9: Model Evaluation
Topics Covered	Accuracy, Precision, Recall, F1, ROC-AUC, Cross Validation
Live Demonstration / Case Study	Compare ML models
Hands-On Activity	Evaluate performance metrics
Outcome	Participants gain practical implementation experience

Session	Session 10: Real-World Data Science Use Cases
Topics Covered	Churn Prediction, Forecasting, Fraud Detection, Recommendation Systems
Live Demonstration / Case Study	Business Solution Mapping
Hands-On Activity	Select right analytical strategy
Outcome	Participants gain practical implementation experience

Session	Session 11: Deploying Data Science Solutions
Topics Covered	FastAPI, Prediction APIs, Docker Basics, Model Packaging
Live Demonstration / Case Study	Deploy ML APIs
Hands-On Activity	Expose prediction endpoints
Outcome	Participants gain practical implementation experience

Session	Session 12: Industry Best Practices & Future Trends
Topics Covered	MLOps, Monitoring, Responsible AI, Generative AI
Live Demonstration / Case Study	Enterprise AI Challenges
Hands-On Activity	Discuss future trends
Outcome	Participants gain practical implementation experience

4. Final Capstone Challenge

Challenge Title: Customer Churn Prediction Challenge

- Analyze telecom customer datasets.
- Perform data cleaning and preprocessing.
- Engineer features for machine learning.
- Train and evaluate prediction models.
- Present business findings and recommendations.

Team Size: 3–5 Participants

Duration: Integrated Throughout Day 3

5. Assessment Framework

Assessment Component	Weightage
Hands-On Labs	30%
Knowledge Checks	20%
Case Study Activities	20%
Final Capstone Presentation	30%

6. Expected Deliverables

- Data Cleaning Notebook
- Feature Engineering Workbook
- Machine Learning Model
- Model Evaluation Report
- Business Recommendation Summary

7. Business Outcomes

- Automate repetitive analytical tasks using Python.
- Improve enterprise data quality and reporting accuracy.
- Develop predictive machine learning solutions.
- Support AI and enterprise analytics initiatives.
- Apply industry-standard development practices in projects.

8. Training Approach

Theory: 40% **|** **Hands-On Practice:** 50% **|** **Discussion & Case Studies:** 10%